

Original Report

Interrater reliability of a near-miss risk index for incident learning systems in radiation oncology



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Abstract

Purpose: Tools for assessing the severity and risk of near-miss events in radiation oncology are few and needed. Recent work has described guidelines for the use of a 5-tier near-miss risk index (NMRI) for the classification of near-miss events. The purpose of this study was to assess the reliability of the NMRI among users in a radiation oncology department.

Methods and materials: Reliability of the NMRI was assessed using an online survey distributed to members of a radiation oncology department. The survey contained 70 events extracted from the department's incident learning system (ILS). Survey participants rated each event using the NMRI guidelines, reported their attendance to weekly ILS meetings (used as a surrogate for familiarity with the ILS), and indicated their familiarity with the radiation oncology workflow. Interrater reliability was determined using Krippendorff's alpha. Use of the NMRI to rate actual events during 5 weekly ILS meetings was also assessed and interrater reliability determined.

Results: Twenty-eight survey respondents represented a wide variety of care providers. Krippendorff's alpha was calculated for the whole respondent cohort to be 0.376, indicating fair agreement among raters. Respondents who had the most participation at ILS meetings ($n = 4$) had moderate agreement with an alpha of 0.501. Interestingly, there were significant differences in reliability and median NMRI scores between professions. NMRI use during weekly NMRI meetings (80 events rated), participants showed moderate reliability ($\alpha = 0.607$).

Conclusions: Using the NMRI guidelines, raters from a wide variety of professions were able to assess the severity of near-miss incidents with fair agreement. Those experienced with the ILS showed better agreement, and higher agreement was seen during multidisciplinary ILS meetings. These data support the use the indices such as the NMRI for near-miss risk assessment in patient safety and prioritization of process improvements in radiation oncology.

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Introduction

As with other areas in medicine, the field of radiation oncology has developed increased interest in patient safety and quality improvement. With this interest has come the implementation of incident learning systems (ILS) across many radiation oncology practices throughout the United States and internationally. Guidelines for the development of ILS have been developed and advocated by the American Society for Radiation Oncology and the International Atomic Energy Agency.^{1,2} In the United States, an ILS has been initiated by American Society for Radiation Oncology and the American Association of Physicists in Medicine in the form of the Radiation Oncology-ILS. Radiation Oncology-ILS allows any radiation oncology practice to submit patient safety events to an online database for multi-institutional tracking of events and process improvement.^{3,4} Internationally, the safety in radiation oncology learning system supported by the International Atomic Energy Agency is available with similar objectives.⁵

The implementation of ILS has led to a change in the perception of patient safety from not only addressing events that reach patients, but also focusing on safety events that have the potential to cause harm, but are detected before reaching a patient (ie, near-miss events³). By collecting near-miss event data into ILS, cause-effect relationships can be garnered (eg, through failure modes and effects analysis, FMEA) and process improvements can be implemented to improve patient safety.⁶

Severity scoring systems for ILS have been developed in radiation oncology and other fields. Scoring allows prioritization of patient safety events and subsequent process improvements.^{3,7,8} Published scoring systems,

including the Agency for Healthcare Research and Quality (AHRQ) Harm Score and French Nuclear Safety Authority Scale, measure severity of events from near-miss/no direct patient harm up to direct patient harm and/or death. As attention toward quality and safety increases, the collection and analysis of near-miss events has increased in importance.⁷ However, until recently, there have been limited tools to specifically assess the severity and/or risk of near-miss events. There is a wide spectrum of near-miss reports, from a wrong location almost treated to a minor issue of miscommunication. Having a reliable system to prioritize near-miss events is essential for extracting the most value from ILS programs. Recently, our group reported its experience with an ILS and proposed a Near-Miss Risk Index (NMRI) for the rating of near-miss events. A need was recognized to evaluate interobserver variability of the NMRI and is the basis of this study.⁹

Methods and materials

ILS and NMRI design

The ILS at our institution has been previously described.¹⁰ In the course of implementing our institutional ILS, a 5-tier severity system was devised to rate near-miss events.⁹ Note that the “near-miss events” included here also encompass the concepts of “unsafe condition” or “process improvement” as described by AHRQ taxonomies.⁶ Initially, the process was driven by group consensus and there were no formal guidelines.

Table 1 NMRI guidelines^a

NMRI	Criteria
0 - No potential harm	Event does not pose downstream risk in workflow. or Event is not related to patient safety or quality of treatment.
1 - Mild potential harm	Event may enhance the risk of other downstream errors. or Event may cause emotional distress or inconvenience to patient with no clinical impact.
2 - Moderate potential harm	Event enhances the risk of other critical downstream errors. or Temporary pain or discomfort for patient. or Deviations from best practices, but with no obvious clinical impact.
3 - Severe potential harm	Limited barriers to prevention of problem or Event with potential clinical impact that is noncritical.
4 - Critical potential harm	Extremely limited barriers to prevention of problem. or Event with potentially critical clinical impact.

NMRI, near-miss risk index.

^a See Ref. 9 for additional description and case examples.

Table 2 Examples of survey events and NMRI scoring

Event description	NMRI score	Number of raters
Event 59: "Wrong dataset used on replan."		
Dosimetrist: "Rescanned patient to see if tumor volume has shrunk. Primary data set for replan was incorrectly set to original CT data set acquired 3 months prior not new scan."	0	0
	1	2
	2	5
	3	10
	4	10
Event 26: "Late work into dosimetry."		
Dosimetrist: "Patient simulated day 1, 'Ready-to-plan' notification request arrived to dosimetry the morning of day 29 for IMRT film day 31, treat day 34. Had to stop what I was doing to get plan to physics ASAP."	0	3
	1	9
	2	5
	3	4
	4	1
Event 14: "Patient delayed by radiology department."		
Patient care coordinator: "Patient's family member upset with radiology...found out that 3 patients were running late there. The family felt overbooking is a problem here and stated they would be going to an outside hospital next time. Family member works at a hospital and is rather irritated. I gave patient full paid parking."	0	13
	1	11
	2	0
	3	0
	4	0

CT, computed tomography; IMRT, intensity modulated radiation therapy; NMRI, near-miss risk index.

However, as near-miss event data were collected, the need for specific guidelines was realized and these were created in the form of a NMRI that rates events from 0 (no risk) to 4 (severe risk)

(Table 1) to rapidly assess event severity and allow prioritized process improvements.⁹ The guidelines are focused largely on the potential severity of events, but the scale also includes the concept of detectability similar to FMEA per American Association of Physicists in Medicine TG-100; therefore, the term "risk index" was adopted.¹¹

Survey-based and real-world NMRI interrater reliability

Interrater reliability of the NMRI was assessed using an online survey designed in REDCap (Vanderbilt University, Nashville, TN)¹² sent to all members of the radiation oncology department involved in the patient care workflow, including patient care coordinators, information technology specialists, radiation therapists, nurses, physicians, physicists, engineers, and dosimetrists. Participants anonymously reported their professional role, frequency of attendance of weekly ILS meetings, frequency of near-miss event reporting, and relative understanding of the radiation oncology workflow (Table E1 available as supplementary material online only at www.practicalradonc.org). The remainder of the survey included 70 selected near-miss events taken from actual near-miss events recorded in the departmental ILS database and edited for clarity. Event text was edited to remove identifiers, typographical errors, and extraneous information. Survey items were selected to represent a wide variety of situations, with the only criteria being that they represent an even distribution of previously assessed NMRI (14 events from each NMRI category) and that sufficient information was documented in the ILS to allow

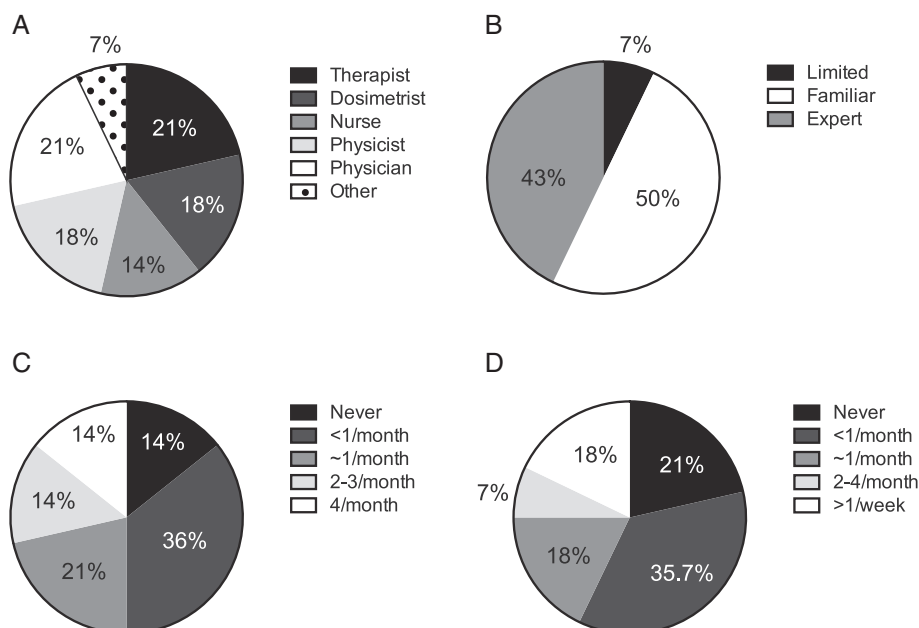


Figure 1 Characteristics of survey respondents. Respondents reported their profession (A), familiarity with the radiation oncology workflow (B), frequency of incident learning system meeting attendance (C), and frequency of near-miss event reporting (D).

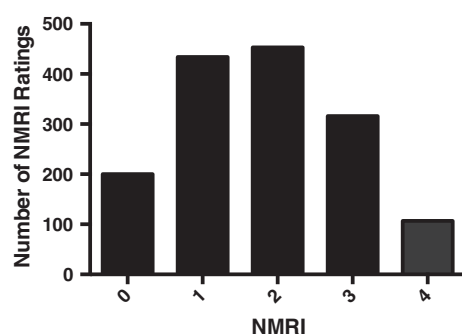


Figure 2 Distribution of near-miss risk index (NMRI) scores by all respondents. The distribution of NMRI scores for all rated survey events (n= 1510) is displayed.

determination of an NMRI. Participants were asked to rate each event according to the NMRI guidelines, which were provided as a downloadable text document. No formal training was provided. Participants could return to the survey later if interrupted before its completion. Some survey respondents (n=7) initiated the survey and completed the profession and ILS participation questions but did not complete any near-miss event items. These responses were excluded from any subsequent analysis.

Examples of 3 survey near-miss events are shown in Table 2, such as event 26, in which delayed submission of planning structures to dosimetry caused a disruption in workflow. This event, although not itself a safety-critical event, leads directly to rushing, which is a causal factor in many error scenarios.

We also assessed the interrater reliability of the NMRI among participants scoring events reported and analyzed during 5 weekly ILS meetings. During meetings, the multidisciplinary group discusses events, gathers additional information, and comes to a consensus regarding causal factors and NMRI. Individual scoring was assessed in real time using PollEverywhere.com, an Internet-based polling software that is accessible by PC, smartphone, and tablet. Participants were asked to anonymously rate events using the NMRI before the group came to a consensus about the NMRI score for a given event.

Statistical analysis

Descriptive statistics including median, mean, range, interquartile range, and standard error of the mean were

computed for NMRI scores among the whole cohort and various subgroups. Because the NMRI is an ordinal system and the distribution of NMRI scores failed D'Agostino and Pearson omnibus normality testing ($P < .0001$), nonparametric tests were used. For comparisons of NMRI scores between 2 groups, the Mann-Whitney U test was used; for comparing 3 or more groups, the Kruskal-Wallis test was used. All tests were 2-tailed, and $\alpha \leq 0.05$ was used as the criterion for statistical significance.

Interrater reliability was determined using Krippendorff's alpha, which allows ordinal rating systems (eg, NMRI 0-4) and missing data.¹³ For the purposes of this study, the scale proposed by Landis et al was appropriated for interpretation of the alpha statistic as follows: <0.00, poor agreement; 0.00-0.20, slight agreement; 0.21-0.40, fair agreement; 0.41-0.60, moderate agreement; 0.61-0.80, substantial agreement; and 0.81-1.00, almost perfect agreement.¹⁴ Interrater reliability was tested for the whole cohort as well as subgroups defined by profession (eg, radiation therapist, physicist) and reported ILS meeting attendance and familiarity.

Results

Out of 102 departmental staff surveyed, there were 28 respondents with analyzable data representing a wide variety of care providers (Fig 1A). Most respondents reported reasonable familiarity with the radiation oncology workflow (Fig 1B). There was a balance between regular ILS meeting participants and nonparticipants (Fig 1C) and also between rare/never submitters of near-miss events and those who submitted reports more frequently (Fig 1D).

Seventeen of the 28 (60%) respondents rated all 70 events. The total number of rated events was 1510; the median number of rated events was 60. The distribution of NMRI scores for all events in the survey is shown in Fig 2. Some events had a wide range of NMRI scores, whereas others were more limited. For example, events such as 26 (Table 2) were rated as high as 4 and as low as 0, whereas all ratings for event 14 were 0 or 1. Thirteen events had an NMRI range of 4, indicating the highest variability in scoring (e-Table 2). Although diverse in nature, a common feature in these events is delayed work and/or events that are potentially critical but have a high likelihood of discovery.

The interrater reliability of the NMRI among respondents was assessed (Table 3). For the entire respondent cohort, Krippendorff's alpha=0.376, indicating fair agreement among raters. We hypothesized that raters who regularly participated in weekly ILS meetings would have higher agreement because of more familiarity with the NMRI and the departmental workflow in general. For survey respondents attending on average 2 or more meetings per month (n=8), alpha increased to 0.430.

Table 3 Interrater Reliability of the NMRI by ILS Participation

Group	Number of raters	Krippendorff's alpha
Entire respondent cohort	28	0.376
Attended ILS meetings 2-3 times per month	8	0.430
Attended ILS meetings 4 times per month	4	0.501

ILS, incident learning system; NMRI, near-miss risk index.

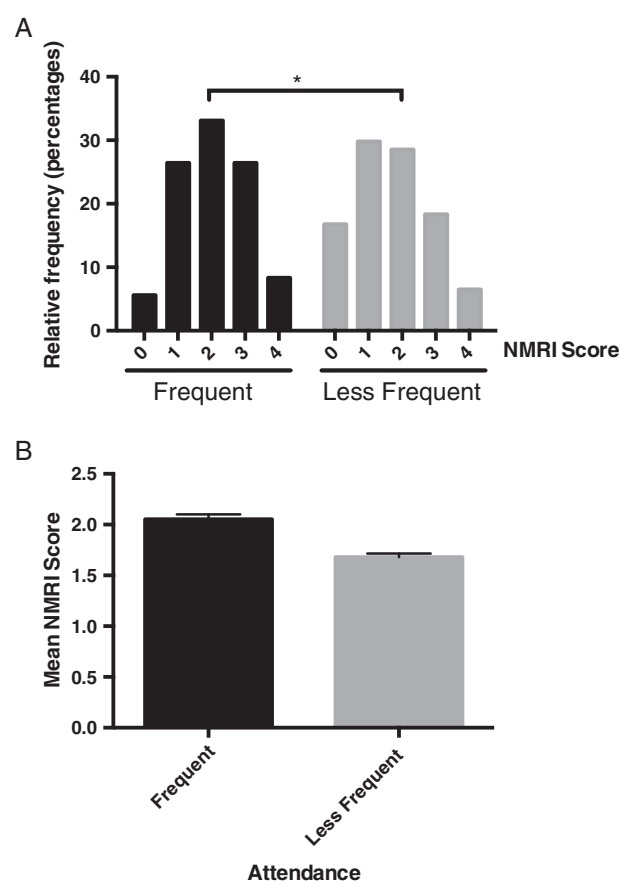


Figure 3 Distribution of near-miss risk index (NMRI) scores between frequent and less-frequent incident learning system (ILS) meeting attendees. (A) The relative frequency of NMRI scores 0-4 were determined for respondents reporting ILS meeting attendance 3-4 times per month (frequent) versus 2 or fewer times per month (less frequent). The number of events scored for each subgroup was 480 and 1030, respectively. Median NMRI score for both groups was 2. Distributions were compared by Mann-Whitney test, *** $P < .0001$. (B) The mean NMRI scores of frequent and less frequent ILS meeting attendees are displayed. Error bars represent standard error of the mean.

Those who reported attending almost every meeting ($n = 4$) had even higher agreement with an alpha of 0.501, indicating moderate agreement. The distribution of scores among frequent attendees versus less-frequent attendees was compared, revealing statistically significant differ-

ences in the score distributions by Mann-Whitney test ($P < .001$, Fig 3A). Less-frequent attendees had lower mean scores (Fig 3B), whereas the median NMRI for each group was 2.

We hypothesized that different professional groups might have different levels of agreement. As demonstrated in Table 4, physicians and dosimetrists had the greatest degree of interrater agreement, whereas nurses and radiation therapists had the least intragroup agreement. We also found significant differences in NMRI score distribution between raters of separate professions (Fig 4).

Finally, we assessed real-world use of the NMRI during 5 weekly ILS meetings. Eighty events were scored among participants for a total of 549 responses. Interrater reliability was higher ($\alpha = 0.607$) than that reported by individual survey participants discussed previously.

Discussion

Incident learning systems allow the collection and analysis of patient safety events for safety improvement efforts. The ability to properly classify and rate the importance of patient safety events is crucial to allow proper prioritization of process improvements. As the use of ILS becomes more widespread, the need for reliable risk assessment tools has become apparent. Several patient safety scores have been used in radiation oncology.^{3,7,8,15-17} Most of these include patient safety events and near-miss events on the same scale, with several having only 1 category for near-miss events.^{3,7,8,15} As a result, granularity in risk assessment for near-miss events is suboptimal. The goal of this study was to assess the reliability of a proposed system specifically designed for prioritizing near-miss events.

This study supports the use of risk assessment systems such as NMRI for ILS, although there should be continued effort for improving interrater reliability. Overall interrater reliability using the survey was fair and was higher for those who had frequent ILS participation (Table 3). Even greater reliability was seen during actual ILS meetings. Active participation in ILS is expected to provide additional perspective and enhance understanding of the potential risks of near-miss events to the overall workflow outside of profession-specific areas. The reliability of the NMRI appears comparable to other patient safety scoring systems. One of the most widely used is the AHRQ Harm Score, a categorical severity score of patient injury. Williams et al tested the interrater reliability of the Harm Score among participants in the University Healthcare Consortium Patient Safety Net¹⁸ and Fleiss' kappa was 0.47 for general respondents and 0.51 for safety managers (experts). Although they have some similarities, it is important to contrast the Harm Score as a tool to rate the severity of explicit patient safety events as opposed to

Table 4 Interrater reliability of the NMRI by profession

Respondent group	Number of raters	Krippendorff's alpha
Nurse	4	0.262
Radiation therapist	6	0.275
Medical physicist	5	0.381
Dosimetrist	5	0.415
Physician	7	0.479

NMRI, near-miss risk index.

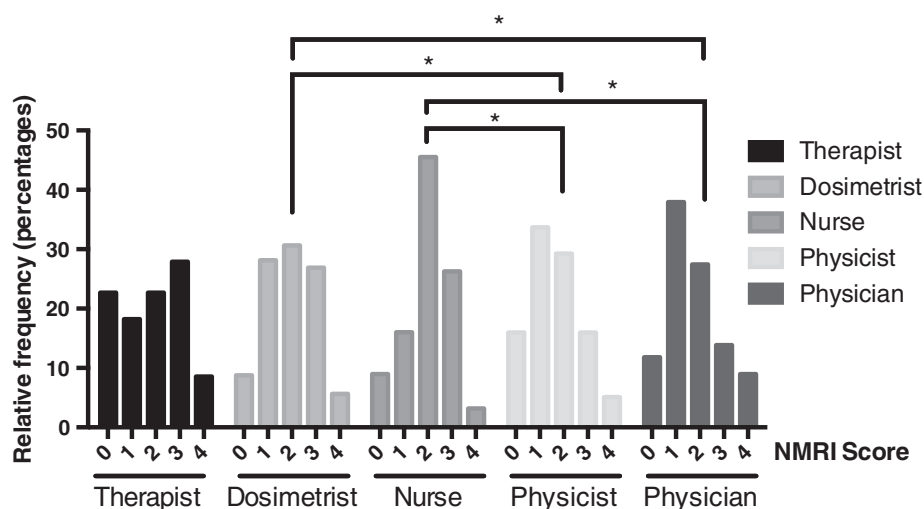


Figure 4 Distribution of near-miss risk index (NMRI) scores for all events by most common professions. (A) The relative frequencies of NMRI scores for each event for therapists ($n = 269$), dosimetrists ($n = 320$), nurses ($n = 156$), physicists ($n = 294$), and physicians ($n = 390$). The median NMRI score for each group as 2. Comparisons were made using Kruskal-Wallis test and Dunn's test for multiple comparisons. $*P < .05$.

the risk of near-miss events in the case of the NMRI. Furthermore, the Harm Score was introduced in 2010 and significant training resources have been invested in its implementation.¹⁸ The differences between Fleiss' kappa and Krippendorff's alpha are also acknowledged. Nevertheless, the NMRI's reliability compares reasonably well with that of the Harm Score, suggesting that a reliable rating system for near-miss events in radiation oncology is achievable with appropriate implementation.

Other groups have analyzed the interrater reliability of ILS information. Kapur et al assessed the reliability of causality factors and FMEA and found a wide range of Fleiss' kappa coefficients depending on the factor measured.¹⁸ For example, high kappas (eg, 0.837) were calculated for various nominal data entered into their institutional ILS. For use of FMEA, the authors reported fair reliability (kappa = 0.533) and good reliability (kappa = 0.629) for assessing severity and detectability of "high-risk processes," respectively, but poor reliability for determining likelihood-of-occurrence (kappa = 0.140). Although there are many caveats to a direct comparison, the reliability of the NMRI is expected to fall in a similar range as the severity and detectability kappas reported by Kapur et al because of the NMRI's design. Nevertheless, the comparable reliability speaks to the accessibility of the NMRI as an assessment tool.

In our study, analysis of the reliability among different professions revealed moderate differences with physicians and dosimetrists having the highest reliability and nurses and therapists the least. Differences in NMRI scores and reliability between professions (Fig 4 and Table 4) may reflect differences in training, experience, familiarity within the ILS, and perspective within the workflow. The latter is important because events often originate in

treatment planning and are discovered in treatment delivery.⁹ Again, increased participation in the ILS is expected to mitigate some of the variability between professions. Furthermore, our data support multidisciplinary risk assessment over individual risk assessment.

Both our study and that of Williams et al demonstrate higher interrater reliability among those with more experience.¹⁷ Additional proactive methods of increasing experience might include additional education, training of staff, and participation in ILS meetings. Broader efforts such as formal guidelines from national consensus groups and practical training at society meetings may also improve the reliability of risks assessments and enhance the quality of ILS data in Radiation Oncology-ILS and across the incident learning community.

Several limitations of this study are recognized. First, many survey items, although taken from real near-miss event reports, contained relatively limited information with which to assess the NMRI. In our system, assessment of risk is performed by the multidisciplinary team through group discussion, which often requires further investigation and gathering multiple perspectives of an event. NMRI assessment is not made until sufficient information is available. Indeed, interrater reliability of NMRI was higher when assessed for participants in real-time ILS event rating, likely because of greater information being available about the event versus the short descriptions in the survey data. As a related issue, regular ILS participants may have recalled some of the events in the survey, thereby improving their ability to rate the event's risk. However, this effect is likely to be small given that, at the time of the survey, the items represented only 2% of event reports in the ILS spanning a 34-month period. Last, although reliability among raters was assessable, reliability

among events was not. More specifically, whether certain events or event types invite higher variability of NMRI scores (e-Table 2) remains undetermined and warrants more investigation.

Conclusions

Risk scores for near-miss events are valuable to prioritize events and direct process improvements. The NMRI is a straightforward tool in assessing the risk of near-miss events to patients and the radiation oncology workflow and achieved fair agreement among general users. Our data suggest that the NMRI's reliability approaches that of well-known indices used for rating explicit medical incidents. This work provides useful insights into the implementation and use of risk assessment tools in radiation oncology ILS. Furthermore, it supports multidisciplinary review of near-miss events as part of a quality ILS. Finally, as more institutions participate in ILS, reliability testing will be an important component of ensuring the quality of ILS data and the consequential improvements in patient safety and care.

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